

Chapter 96 — Acute Care of Urinary Lithiasis

INITIAL PRESENTATION

- **Classic presentation: unilateral severe flank pain** progressing toward the lower abdomen and genitalia. It is not positional — the patient cannot find a comfortable position and writhes — and waxes and wanes, usually described as excruciating. Less commonly it is a dull pressure.
- **A dull-pressure presentation** should raise suspicion of a **more chronic obstruction**.
- **Stone approaching the vesicoureteral junction** → pain at the tip of the urethra, dysuria, persistent frequency and urgency. **Sudden relief of pain** suggests passage into the bladder.
- **Nausea and vomiting in about half** of acute cases.
- **Haematuria (microscopic or macroscopic) in about 90%**, gross in about one-third. **Its absence does not exclude a stone** — more than one in ten proven passages have none.
- **Fever and chills are uncommon**; when present they suggest an **infected obstructing stone** and the need for more urgent intervention.

Office Versus Hospital Setting

- **Most present to urgent care or the emergency department** with classic renal colic; milder cases present to the office — **maintain a high index of suspicion**.
- Office patients seen after the initial colic may report dull pressure or **lower urinary tract symptoms** (dysuria, urgency, frequency), attributable to a calculus that has migrated to the **ureterovesical junction (UVJ)**.

Laboratory Tests

- **Routine tests suffice for most**: serum chemistry and urinalysis; a **complete blood count (CBC) with differential** because systemic infection may be present; a **pregnancy test in every patient of reproductive age**.
- **Results are usually normal and renal function preserved**, unless there is significant dehydration, bilateral obstruction, or an absent or abnormal contralateral kidney.
- **Raised neutrophils and white blood cells (WBCs) may be a stress response**, not necessarily urinary tract infection (UTI).
- **Microscopic haematuria is present in more than 90%** of acute stones, yet **haematuria is a poor predictor of a stone** — sensitive, not specific.
- **WBCs, leucocyte esterase and nitrites** raise suspicion of infection but are **not diagnostic**. **More than 50 WBCs per high-power field** is associated with a positive culture in the majority; **nitrites** increase the likelihood of concurrent UTI.

Imaging Studies

- **Non-contrast computed tomography (NCCT; CT) of abdomen and pelvis is the gold standard**, especially in first-time stone formers. American College of Radiology Appropriateness Criteria: **sensitivity 97%, specificity 95%**, the examination of choice.
- NCCT evaluates kidneys, ureters and bladder, measures **true stone volume, size, location and skin-to-stone distance**, and shows secondary signs — hydronephrosis, hydroureter, perinephric stranding, suspected calyceal rupture.

- **Radiation — ALARA (as low as reasonably achievable)**, the radiation-protection principle since 1977. United States Nuclear Regulatory Commission **occupational whole-body limit 50 mSv/year**. **Standard NCCT delivers roughly 8–16 mSv**, varying widely by protocol and facility; reduced-dose protocols are used in only a small minority of renal-colic studies.
- **Low-dose NCCT (≤ 3 mSv; effective dose about 0.5–2.8 mSv)**: sensitivity and specificity around 90% or better; weak below 2–3 mm and in obesity, but comparable to standard NCCT for stones ≥ 3 mm (meta-analysis).
- **STONE score**: five criteria grouped into low, moderate and high risk, designed to identify uncomplicated ureteral stones, estimate the likelihood of an alternative diagnosis and limit NCCT use. **External validation failed** to show accuracy sufficient to dispense with NCCT; criterion changes have been proposed.

IMAGING MODALITIES IN ACUTE RENAL COLIC

Modality	Role	Strength	Limitation
NCCT	Gold standard ; first-line in first-time formers and the elderly	Stone size, volume, location, skin-to-stone distance; secondary signs; alternative diagnoses	Radiation; misses matrix and indinavir stones
Low-dose NCCT (≤ 3 mSv)	Good alternative; in pregnancy when necessary (after US and MRI)	Comparable accuracy for stones ≥ 3 mm	Weak for stones < 3 mm and in obesity
Dual-energy CT (DECT)	Composition estimate to guide treatment	Attenuation differences at two X-ray energies; identifies uric acid stones (sensitivity and specificity $\geq 95\%$)	Rarely changes acute management
Ultrasound (US)	First-line in pregnancy and children ; in recurrent formers or with recent CT showing true burden	No radiation, cheap, easy; normal US associated with less intervention; outcomes and complications similar to NCCT-first strategy	Uses hydronephrosis as surrogate ; operator dependent; up to 40% still need CT
Plain abdominal X-ray (KUB)	Avoid for acute diagnosis	Most stones calcium based and radiopaque	Lowest sensitivity and specificity ; obesity, bowel gas, other calcifications
Magnetic resonance imaging (MRI)	Second-line in pregnancy and children when US non-diagnostic	Accurately shows secondary effects of obstruction; no radiation	Poor stone detection; costly; time consuming

- The table is testing **which modality answers which question**: when the diagnosis is already established (recurrent former, recent CT), imaging only has to show hydronephrosis, so US suffices; when the diagnosis is unknown, only NCCT gives stone data and alternative diagnoses. Trap: **plain X-ray is not an acceptable acute diagnostic test** despite most stones being radiopaque.

Concept — NCCT answers every acute question at once — size, site, secondary signs and alternative diagnoses; every other modality is a trade of information for radiation, accepted only where radiation matters more (pregnancy, children, repeat imaging of a known stone).

PAIN MANAGEMENT IN RENAL COLIC

Mechanism of renal colic:

Stone enters ureter → obstruction → stretch of renal pelvis and capsule → local prostaglandin release → diuresis and vasodilation → **rising intrarenal pressure**; prostaglandins also act directly on the ureter → **smooth-muscle spasm**.

- **Shared innervation** → pain referred to intestine, groin and genitalia; a **distal ureteral stone refers to the ipsilateral testis, labium or groin**.
- **Prostaglandins (arachidonic acid metabolites) drive the pain and inflammation. Non-steroidal anti-inflammatory drugs (NSAIDs) inhibit cyclo-oxygenase (COX)** and so stop their production, without the respiratory and consciousness depression of opioids; intravenous NSAIDs act immediately, reducing smooth-muscle stimulation and ureteral spasm.
- **Bleeding and renal-function concerns are not clinically significant** at common doses.
- **NSAIDs versus opioids**: comparable efficacy (no difference at 30 and 60 minutes in a large systematic review), with **less rescue analgesia and less vomiting** with NSAIDs; one systematic review found greater pain reduction with NSAIDs.

ANALGESIC OPTIONS IN RENAL COLIC

Agent	Place	Rationale	Notes
NSAID — ketorolac 30 mg, any route	First line (National Institute for Health and Care Excellence [NICE] and European Association of Urology [EAU])	COX inhibition → less prostaglandin, less spasm	Contraindications: renal failure, gastrointestinal bleeding, pregnancy (depending on gestational age)
Intravenous paracetamol (acetaminophen)	Second line	Non-opioid alternative	When NSAIDs are contraindicated or insufficient
Dexamethasone 8 mg added to ketorolac 30 mg	Possible adjunct	Additional anti-inflammatory effect	Single study: fewer patients needed opioids or antiemetics than with ketorolac alone
Opioids	Reserved	μ-opioid receptor agonism	Intractable pain, or pain unresponsive to NSAID and paracetamol; continue the NSAID — morphine plus ketorolac beats either alone
Sterile water skin injection	When all the above are unavailable or contraindicated	Described since 1949; four injections around the painful area to a single trigger-point injection	Systematic review of randomised controlled trials (RCTs): better than oral paracetamol, equivalent to diclofenac and morphine
Desmopressin (DDAVP)	Investigational	Less urine production → rapid fall in intrarenal pressure	Pain reduction in one emergency-medicine study
Antispasmodics	Not effective	—	No demonstrated efficacy (NICE)
High-volume fluids / diuretics	Not for analgesia or passage	—	No supporting data; fluids only to treat vomiting and dehydration

- The pattern: agents acting on **prostaglandins or intrarenal pressure** work; agents aimed at spasm alone (antispasmodics) or at "flushing" the stone (fluids, diuretics) do not. Traps: **forced fluids to aid passage**, and **stopping the NSAID when an opioid is started**.

OPIOIDS

- **History:** morphine isolated from opium 1806; chemical formula deduced 1847; hypodermic needle 1853 → widespread clinical use.
- **Three classical receptors:** δ (DOR), κ (KOR) and μ (MOR); MOR is the most clinically relevant. Morphine is the reference MOR agonist and also acts on the other two; opioids remain the **benchmark against which all analgesics are measured**.
- **Adverse effects:** reduced consciousness, respiratory depression, constipation, nausea and vomiting, **urinary retention**, pruritus, muscular rigidity, miosis, dysphoria; **euphoria underlies abuse potential**.
- **Opioid crisis:** most United States overdose deaths involve an opioid, quadrupled since 1999; **United States crisis declared 2017**, with public-health emergencies declared elsewhere.
- **Dependency in opioid-naïve patients rises with each additional day prescribed from day 3, most sharply after day 5.** Highest long-term abuse probability with **long-acting opioids**, then tramadol. Younger patients (<45 years) given two prescriptions rather than one roughly double their risk.
- **Stewardship:** multimodal non-opioid protocols cut opioid prescribing at the cost of more admissions and repeat emergency visits; surgeon-level protocols did not change other prescribers' habits → **system-wide stewardship needed**.
- **Current recommendation: opioids only for intractable pain or pain unresponsive to NSAIDs and paracetamol, with the NSAID continued.**

BOX 96.1 QUESTIONS FOR THE EMERGENCY DEPARTMENT CONSULTING PHYSICIAN

Domain	Items
Vital signs	Fever; blood pressure; tachycardia; respiratory rate; altered mental status
Stone variables	Size; location; hydronephrosis; multiple stones
Laboratory studies	CBC; blood urea nitrogen and creatinine; urinalysis
Clinical course	Pain level and response to initial treatment; ability to keep fluids down; nil-by-mouth status
Existing conditions	Elderly and frailty; immunocompromise; diabetes
Previous history	Previous surgery for acute stone disease; recurrent UTIs

- The box is a **triage instrument**: vital signs and laboratory results screen for sepsis and renal compromise, stone variables predict passage and stent success, and host factors lower the threshold for drainage — together they decide **discharge with trial of passage versus urgent decompression**.

Concept — Renal colic is a prostaglandin-mediated pain, so COX inhibition treats its cause while an opioid only masks it; hence NSAIDs lead, opioids are added rather than substituted, and the dependency signal — tied to prescription length and formulation — argues for the fewest days possible.

MEDICAL EXPULSIVE THERAPY

- **Medical expulsive therapy (MET):** medication to facilitate passage of a ureteral stone into the bladder; **α -blockers are the principal agents**.
 α -Blocker mechanism: less ureteral spasm and lower force and frequency of peristalsis → larger urine bolus per peristaltic wave → **higher pressure proximal to the stone**.
- **NSAIDs alone** reduce inflammation and oedema but **do not improve passage**.
- **American Urological Association (AUA) and EAU:** MET for distal ureteral stones <10 mm. Tamsulosin is the most studied; other α -blockers offer the same benefit.

EVIDENCE FOR α -BLOCKER MET

Study	Population / comparison	Verdict
SUSPEND — large multicentre RCT (2015)	1136 patients, single ureteral stone; tamsulosin vs placebo	No difference in need for intervention at 4 weeks (19% vs 20%)
Double-blind RCT	Stones up to 9 mm; 4 weeks tamsulosin vs placebo; 90-day follow-up	Similar passage rates
Double-blind multicentre RCT	Stones ≤ 10 mm	No overall difference; benefit in 5–10 mm subgroup ; little or none < 5 mm
Large double-blind multicentre trial	Distal ureteral stones	Benefit for distal stones > 5 mm
Meta-analysis of RCTs (tamsulosin)	Tamsulosin vs control	Higher and faster expulsion, less colic, less surgery; benefit confined to > 5 mm , none ≤ 5 mm
Meta-analysis of RCTs (tamsulosin and silodosin)	Ureteral stones by location	No benefit proximal; clear benefit distal ; silodosin superior to tamsulosin
Meta-analysis	Silodosin vs tamsulosin	Silodosin: higher and faster expulsion

TAMSULOSIN VERSUS SILODOSIN

Tamsulosin	Silodosin
Most studied α -blocker for MET	Superior expulsion rate and time in meta-analyses
Lower rate of retrograde ejaculation	Retrograde ejaculation roughly twice as frequent

- The trials apparently conflict only until stratified by **size and site**: negative trials enrolled all sizes and sites, while benefit clusters in **distal stones of 5–10 mm**. Trap: **SUSPEND does not abolish MET** — it was an unstratified population.
- **Recommendation: offer and discuss MET for a ureteral stone < 10 mm when immediate intervention is not required. A trial of passage, with or without MET, for 4–6 weeks, after which intervention should be considered.**

Concept — α -Blockade helps only a stone large enough not to pass unaided (> 5 mm), small enough to pass at all (< 10 mm) and already distal; outside that window it adds nothing, which reconciles the negative trials with the positive meta-analyses.

URGENT INTERVENTION

- **Spontaneous passage is not universal**, and depends on **size, location and especially transverse width** rather than length.

SPONTANEOUS PASSAGE BY SIZE, WIDTH AND LOCATION

Determinant	Passage
Size (series before MET)	1 mm 87%; 5 mm 60%; 10 mm 27% — bands: 1–4 mm 78%, 5–7 mm 60%, 8–10 mm 39%
Location	UVJ 92% (1–4 mm stones); distal 75%; mid 60%; proximal 48%
Width (later study; overall 80% passed, mean 37 days to intervention when needed)	0–3 mm 98%; 4 mm 81%; 5 mm 65%; 6 mm 33%; > 6.5 mm 9%

Determinant	Passage
AUA estimate	Eventual intervention in ~50% of 4–6 mm stones and in the large majority of stones >6 mm

- The table's pattern: passage falls steeply **beyond 5–6 mm width** and from distal to proximal. Trap: **measure width, not length**.
- **AUA indications for urgent intervention: urinary infection with obstruction**; bilateral obstruction; obstruction of an anatomically or functionally solitary kidney; immunosuppression; impending renal deterioration; pain refractory to analgesics; intractable nausea and vomiting.
- **Acute obstruction alone does not threaten long-term renal function unless infected**; progressive deterioration is a concern only with **long-term obstruction (beyond about 2 weeks)**.
- **Decompression: retrograde ureteral stent or percutaneous nephrostomy (PCN)**, chosen by patient and surgeon preference, local practice and availability — ideally a shared decision.

URETERAL STENT VERSUS PERCUTANEOUS NEPHROSTOMY (META-ANALYSIS OF RCTS)

Ureteral stent	Percutaneous nephrostomy
More haematuria and dysuria (urinary and voiding symptoms)	Longer hospital stay ; fewer urinary symptoms
No difference in improvement of septic parameters	
Higher failure with large or multiple stones	Possible advantage with multiple stones or large transverse diameter

PREDICTORS OF STENT-INSERTION FAILURE

Predictor	Threshold
Transverse stone diameter	≥9.5 mm on CT measurement; the chapter's key points round this to >10 mm
Pericalcular ureteric thickness (P-CUT)	≥7.5 mm — strongest predictor
Multiple calculi	Higher failure (septic obstructed cohort)
Larger stone size	Higher failure (septic obstructed cohort)

P-CUT measurement: on CT, measure the widest transverse diameter of the ureter including the stone → measure the stone's maximum transverse diameter at that level → subtract stone width from total → P-CUT (wall oedema around an impacted stone).

- **Acute extracorporeal shock wave lithotripsy (ESWL):** described since 1997, advocated mainly for **proximal ureteral stones**; faster passage and symptom relief with low morbidity; treatment within 24 hours of the first colic may benefit from the **absence of significant impaction**.
- **Acute ureteroscopy:** most successful for **distal stones and stones <10 mm**; complication rate is higher across all sizes than when restricted to <10 mm → offer **only smaller stones, in favourable conditions**, after discussion.
- **Acute ESWL or ureteroscopy only in the absence of infection or its signs**; infection mandates drainage.

Concept — *Infection plus obstruction is the emergency; obstruction alone is not. Drainage — stent or nephrostomy, largely interchangeable except when stone bulk defeats the stent — treats the emergency, while definitive acute lithotripsy is reserved for uninfected, favourable stones.*

SPECIAL CONSIDERATIONS

Pregnancy

- **Parous women under 50 have more than double the stone incidence** of nulliparous women, rising with parity.
- Physiological changes of pregnancy counterbalance one another, so **prevalence of urolithiasis does not increase in pregnancy**; symptomatic stones affect under 1% of pregnancies but carry **adverse birth outcomes** — low birth weight, pre-eclampsia, preterm birth — especially when surgery is needed.
- **Intervention needed in roughly one-fifth to one-third** (about 35% in the chapter's key figure): stents most often, then ureteroscopy, then nephrostomy. **Preterm delivery rises stepwise**: conservative management < procedural intervention < **nephrostomy (highest)**.

Imaging

US first (all guidelines) → MRI second (AUA, EAU) → low-dose NCCT when necessary (endorsed by the American College of Obstetricians and Gynecologists [ACOG]; shared decision, after other modalities).

- **Physiological hydronephrosis in ~90%** (second and third trimesters) makes hydronephrosis unreliable; it **does not usually extend beyond the iliac vessels**, so **distal ureteral dilatation points to a stone**. An **absent unilateral ureteric jet** helps but occurs without obstruction.
- **Intrarenal resistive index (RI)**: mean RI ≥ 0.7 (vs ≤ 0.6) is specific but insensitive; a **difference of ≥ 0.06 from the contralateral kidney** is highly sensitive and specific for obstruction.
- **MRI**: useful positive predictive value despite poor stone visualisation. **Gadolinium** is teratogenic in animals but not in human studies, and usually unnecessary; ACOG: use only where benefit clearly outweighs risk.
- **CT**: rarely first line; **teratogenic risk greatest before or during the first trimester**; fetal abnormality risk **not significant below 50 mSv**; low-dose protocols ≤ 3 mSv, some < 1 mSv — well within the safe range.

Management

- **First line: conservative management**. α -Blockers are insufficiently studied in pregnancy — used by fewer than half of urologists, against near-universal use otherwise.
- **Up to 30% of symptomatic patients need intervention** (ureteroscopy, stent or nephrostomy) — shared decision and individualised plan.
- **Stent vs nephrostomy** by hydronephrosis, preference and **gestational stage**. Both encrust → **change every 4–6 weeks**. Each stent exchange needs anaesthesia, so **the earlier the pregnancy, the more nephrostomy is preferred**. Stents also cause stent colic, **preterm labour soon after placement**, and **early induction in over 40%** for poor stent-pain control.
- **Ureteroscopy**: former advice to wait for the second trimester is superseded — **ACOG Committee Opinion No. 775 (2019)**: no teratogenicity of anaesthetic agents at standard concentrations, no fetal brain effect with exposure under 3 hours; **never deny or delay medically necessary surgery in any trimester**. Meta-analysis: low complication rate, mostly Clavien I–II, no rise in UTI, low preterm labour — lower than stenting in some series → **safe, and avoids repeated tube exchanges**.
- **Fluoroscopy**: minimal exposure, **pelvic shield**, and **C-arm on manual exposure** (automatic mode increases dose) — agree with staff beforehand. Stones in pregnancy are covered further in Chapter 21.

Frailty and the Elderly

- **Same initial management, but atypical presentation is more likely.**

- Mean age at first stone about 40; **a first obstructing stone after 60 is less likely** → broaden the differential to **acute myocardial infarction and aortic aneurysm or dissection**.
- Older patients more often have atypical or no pain, fever, diarrhoea, pyuria, UTI and bacteraemia; more atypical composition including **uric acid**; and a **lower chance of spontaneous passage**.
- **Imaging: always CT** — diagnosis and exclusion of alternatives.
- Comorbidity often **precludes NSAIDs; antiplatelets and anticoagulants** complicate procedures; reduced reserve demands **high suspicion for infection and early drainage**.

Transplant Kidney

- **Rare** — low single-digit incidence of stones and obstruction in allografts.
- **Denervated allograft → obstruction is painless**; missed cases present with hydronephrosis, fever and acute renal failure → **regular follow-up and high suspicion**.
- **Endoscopic management preferred**; if retrograde stenting is difficult, **percutaneous access under US or CT guidance**.

SPECIAL POPULATIONS COMPARED

Population	Presentation	Imaging	Management point
Pregnancy	Typical, but physiological hydronephrosis confounds	US → MRI → low-dose NCCT	Conservative first; nephrostomy early in gestation; ureteroscopy safe; tubes changed 4–6-weekly
Elderly / frail	Atypical ; cardiovascular mimics	CT always	NSAIDs often contraindicated; anticoagulation; early drainage
Transplant kidney	Painless (denervation)	US or CT to guide access	Endoscopic preferred; percutaneous if retrograde fails

Concept — *Each special population removes one assumption of standard care: pregnancy removes free use of radiation (imaging reordered), age removes the typical presentation and safe NSAID use, and denervation removes the pain signal — so in each, vigilance must replace the missing cue.*

CONCLUSION

- **Initial management is individualised**; accurate diagnosis gives prognosis and manages expectations.
- **Ideal: conservative management** — pain control and spontaneous passage; when intervention is needed, **decide swiftly to minimise morbidity**.